GlycoAhead

Anticipating a new high-glucose episode from history, CGM and recorded context

A focused digital-twin proof of concept with a working replay application and patient-separated model evaluation.

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Python / ML / database / backend / frontend / evaluation

The problem and intended healthcare workflow

Problem

• A present glucose value does not show whether a new episode will begin soon.

• Recent glucose patterns and recorded context may support earlier review.

• Warnings must be assessed against false-alert burden and actual lead time.

Intended use

• Clinician or researcher reviewing adult Type 2 diabetes CGM records.

• Inspect an alert with its glucose trend, history, meals and medication context.

• Support review; no automated dosing, diagnosis or treatment recommendation.

A precise event, rather than an undefined 'spike'

Prediction contract

- 13 contiguous readings cover the preceding three hours.
- Predict a new onset at +15, +30, +45 or +60 minutes.
- Output is event probability, not an exact future glucose value.

Endpoint and confirmation

- Two readings >180 mg/dL, preceded by two readings <=180 mg/dL.
- Onset is the first high reading; the latest two inputs must be <=180.
- Full outcome confirmation requires readings through +75 minutes.

Illustration: an alert at 12:00 and an onset at 12:30 provide 30 minutes of warning, not 60.

Dataset: linked observations from the same people

ShanghaiT2DM v5

- 100 unique patients; 109 recordings.
- 112,462 retained numeric CGM readings at native 15-minute cadence.
- Linked clinical summary and timestamped meal/medication entries.
- Variable recording lengths; no invented padding.

Provenance and boundaries

- Recording identifiers link history and sensor data.
- Source hashes, quality checks and original timestamps are retained.
- No unrelated EHR-to-wearable joins.
- Chinese cohort; no Indian external validation.

Sources: Figshare dataset 21600933, version 5; Scientific Data, doi:10.1038/s41597-023-01940-7.

Two data streams, 54 deployed predictive inputs

History and glucose

• Age and diabetes duration; assumed available at enrollment.

• 13 glucose lags; mean, variability, minimum and maximum.

• 15- and 60-minute slopes, prior high fraction and time of day.

• Undated labs and outcome summaries excluded.

Recorded context

• Meal counts, timing, food categories and documented portions.

• Medication entries, timing and some documented doses.

• Estimated carbs, protein, fat, fibre and energy.

• Nutrition coverage and assumption indicators.

Food references are estimates, not measured intake

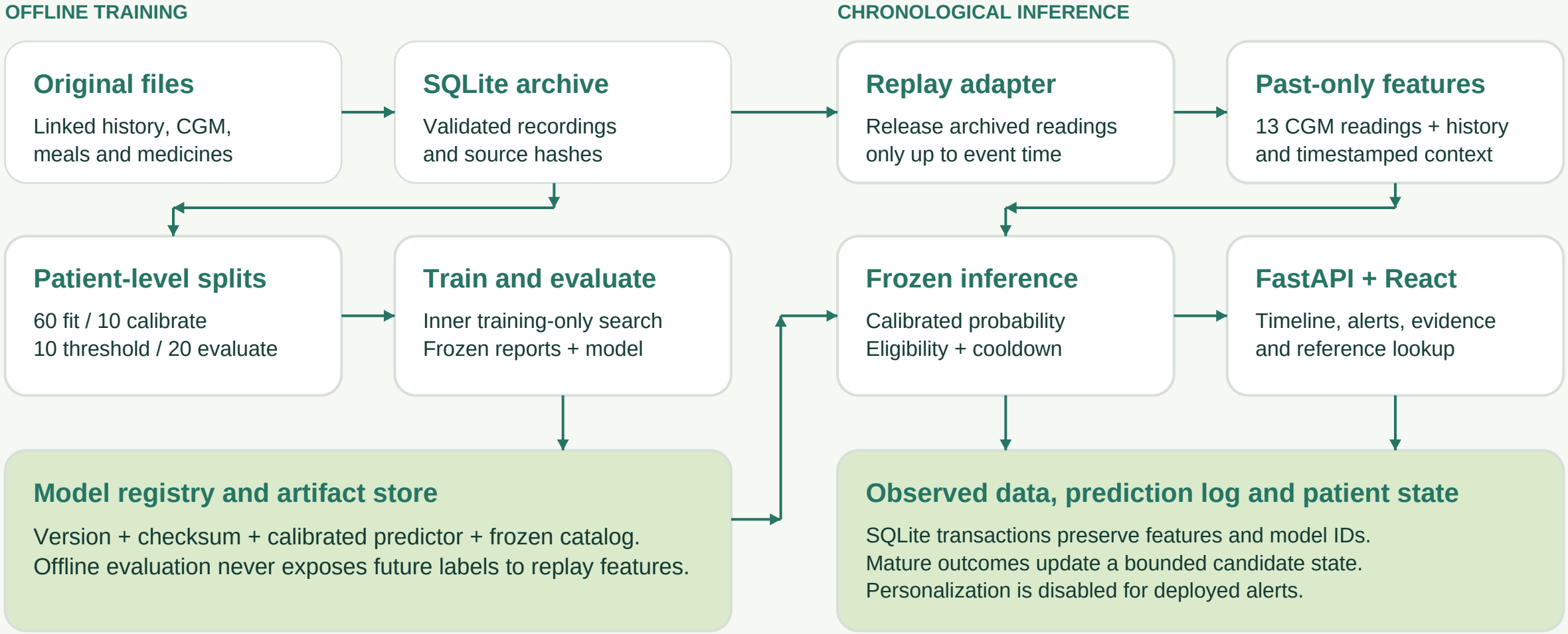
Nutrition pipeline

- Match a recorded food to a preparation-specific reference.
- Scale per-100-g nutrients to the documented edible amount.
- Use explicit generic recipe assumptions for supported mixed dishes.
- Freeze catalog contents with the model version.

Missingness and medication

- Unknown food or quantity remains missing, not zero.
- Incomplete meal totals are not silently presented as complete.
- Medication features describe documented context.
- No insulin-on-board estimate or causal dosing simulation.

Architecture: one app, two execution paths



One local FastAPI worker serves the API and built UI. Training runs as a separate CLI job. No live wearable connection.

Training and tuning without patient overlap

One outer assignment

• 60 patients fit the model; 10 calibrate probabilities.

• 10 select alert thresholds; 20 evaluate performance.

• All visits from one patient remain in the same group.

• Three predefined patient assignments examine robustness.

Training-only selection

• Histogram gradient boosting with sigmoid calibration.

• Eight fixed settings; three inner folds within the 60 training patients.

• Rank by inner patient-weighted average precision.

• Refit, calibrate, select threshold and evaluate separately.

Digital twin: an evolving predictive state

Implemented lifecycle

- New patient starts with population predictions.
- Only matured outcomes update a bounded patient risk offset.
- Cold start: at least 20 spaced outcomes and 24 hours.
- State resets at gaps and recording boundaries.

What is established

- The state-update and provenance mechanisms are implemented.
- Base model weights stay frozen during replay.
- Personalized candidate is visible alongside population risk.
- Personalization benefit is not established; population alerts remain active.

Working product: five connected views

Inspect the patient

- Patient workspace: replay, glucose and probability timelines.
- Meals, medicines, patient state and alert acknowledgement.
- Inspect prediction: saved features, model and state.
- Outage simulation: withhold forecasts until history recovers.

Inspect the system

- Patient cohort: linked recordings, lengths and split membership.
- Model evidence: selected model and repeated comparisons.
- Food reference: nutrients, mappings and source metadata.
- How it works: workflow plus documented API.

Recorded replay is clearly distinguished from a connected live wearable.

Results: compare recall, alert burden and warning

Three-run mean	Recall	False alerts/day	Mean warning
Tuned current (54 features)	80.60%	1.08	26.48 min
Tuned meal timing (72 features)	81.94%	1.09	26.83 min
Tuned pre-meal glucose (60 features)	79.84%	1.01	26.51 min
Simple glucose + trend baseline	81.06%	1.17	24.67 min

Deployed seed-2026 model: 243/294 events caught (82.65%); 1.39 false alerts/day; 30-minute median warning. This single-run result is different from the three-run mean.

Warning times concern caught episodes. Runs share patients. Previously inspected cohort: internal robustness, not external validation. Recall is not overall classification accuracy.

Engineering and reproducibility

Implemented

• FastAPI + React/TypeScript; one local worker.

• SQLite, SQLAlchemy and Alembic migrations.

• Versioned model/catalog, checksums and prediction snapshots.

• Health endpoint, request logs and stale-request handling.

Repository evidence

• Pinned dependencies and local setup/start scripts.

• Existing split/meal-chronology tests and CI build definition.

• Architecture, model card and aggregate comparison reports.

• Docker files included; Docker deployment not yet validated.

Limits that define the next evaluation

Data and model

- Short recordings and only 100 participants.
- Uncertain availability timing for some clinical/context fields.
- Estimated nutrition and potentially missing meal entries.
- The deployed model exceeds the proposed <=1 false alert/day target.

Evidence still required

- External validation, including Indian population data.
- Clinician review and prospective usefulness assessment.
- Demonstrated benefit from patient personalization.
- No clinical safety, production scale or compliance claims.

Future scope and delivery

Next technical steps

- Broader, longer-duration patient data.
- Live CGM adapter with duplicates, late data and gap handling.
- Richer reliably timestamped activity, sleep and treatment context.
- Controlled evaluation before enabling personalized alerts.

Submission deliverables

- Working application and source repository package.
- Architecture PDF and this presentation.
- Selected calibrated model and aggregate evidence.
- Video recorded; unlisted YouTube link being added to README.

Goal: a useful, inspectable forecasting workflow that can be evaluated and extended.

References, licensing and project identity

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Dataset: Shanghai diabetes dataset, v5

https://doi.org/10.6084/m9.figshare.21600933.v5

Original data descriptor: Scientific Data (2023)

https://doi.org/10.1038/s41597-023-01940-7

Project evidence: source code and frozen aggregate reports

README.md; docs/MODEL_CARD.md; docs/TUNING.md; release/selected_model/comparison.json

Code: MIT. Shanghai data: CC BY 4.0. Nutrition references and dependencies retain their separate terms; code licensing does not relicense third-party data.